

Richness plots

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```
# libraries ----
library(tidyverse)

## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.2.1      v readr      2.2.0
## v forcats    1.0.1      v stringr   1.6.0
## v ggplot2    4.0.3      v tibble    3.3.1
## v lubridate  1.9.5      v tidyr     1.3.2
## v purrr      1.2.2
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors

library(readxl)

# data ----

data <- map(1:5, \(x) {
  #read all three taxa
  read_excel("data/Data2024.xlsx", sheet = x) |>
  #pivot
  pivot_longer(-1, names_to = "species") |>
  #rename first column
  rename(plot = 1)
})

#keep copy with separate layers
data1 <- data

#first three sheets are vascular plants, join them together
data <- bind_rows(data[1:3]) |>
  list() |>
  c(data[4:5])

data <- data |>
  #bind list to single df
  bind_rows(.id = "taxon") |>
  #rename IDs
  mutate(taxon = fct_recode(taxon,
```

```

        "Vascular plants" = "1",
        "Saproxylic fungi" = "2",
        "Epiphytic lichens" = "3") |>
  fct_relevel("Saproxylic fungi", after = 2),
  #extract genus
  genus = word(species, 1),
  #extract epithets
  epit = word(species, 2),
  #extract site
  site = substr(plot, 1, 2) |>
  fct_recode("Prati di Tivo" = "PR",
            "Incodaro" = "IN",
            "Venaquaro" = "VE")
) |>
arrange(taxon, species) |>
#these were wide matrices, keep only observed entries
filter(value > 0)

#repeat on layer data
datal <- datal |>
  #bind list to single df
  bind_rows(.id = "taxon") |>
  #rename IDs
  mutate(taxon = fct_recode(taxon,
                            "Tree" = "1",
                            "Shrub" = "2",
                            "Herb" = "3",
                            "Saproxylic fungi" = "4",
                            "Epiphytic lichens" = "5") |>
    fct_relevel("Saproxylic fungi", after = 4),
    #extract genus
    genus = word(species, 1),
    #extract epithets
    epit = word(species, 2),
    #extract site
    site = substr(plot, 1, 2) |>
    fct_recode("Prati di Tivo" = "PR",
              "Incodaro" = "IN",
              "Venaquaro" = "VE")
) |>
arrange(taxon, species) |>
#these were wide matrices, keep only observed entries
filter(value > 0)

#distinct species names
distinct(data, species) |> pull() |> sort()

```

```

##   [1] "Abies alba"           "Acer campestre"
##   [3] "Acer opalus"          "Acer platanoides"
##   [5] "Acer pseudoplatanus"  "Actaea spicata"
##   [7] "Adenostyles australis" "Adoxa moschatellina"
##   [9] "Aegopodium podagraria" "Agrimonia eupatoria"
##  [11] "Agrostis capillaris"  "Ajuga reptans"

```

## [13]	"Allium ursinum"	"Anaptychia ciliaris"
## [15]	"Anaptychia crinalis"	"Aquilegia dumeticola"
## [17]	"Arabis alpina"	"Arabis collina"
## [19]	"Arabis hirsuta"	"Arctium lappa"
## [21]	"Arctium minus"	"Aremonia agrimonoides"
## [23]	"Armillaria sp."	"Arthonia didyma"
## [25]	"Arthonia radiata"	"Arum maculatum"
## [27]	"Asperula laevigata"	"Asplenium ruta-muraria"
## [29]	"Asplenium trichomanes"	"Astragalus glycyphyllos"
## [31]	"Atadinus alpinus"	"Athallia cerinelloides"
## [33]	"Athallia pyracea"	"Atropa bella-donna"
## [35]	"Avenella flexuosa"	"Bertia moriformis"
## [37]	"Biscogniauxia nummularia"	"Bjerkandera adusta"
## [39]	"Brachypodium rupestre"	"Brachypodium sylvaticum"
## [41]	"Bromopsis ramosa"	"Buellia griseovirens"
## [43]	"Bupleurum falcatum"	"Caloplaca cerina"
## [45]	"Caloplaca sp."	"Calycina citrina"
## [47]	"Campanula glomerata"	"Campanula persicifolia"
## [49]	"Campanula rapunculus"	"Campanula tanfanii"
## [51]	"Campanula trachelium"	"Candelariella faginea"
## [53]	"Cardamine bulbifera"	"Cardamine enneaphyllos"
## [55]	"Cardamine graeca"	"Cardamine heptaphylla"
## [57]	"Cardamine kitaibelii"	"Carduus sp"
## [59]	"Carex digitata"	"Carex divulsa"
## [61]	"Carex flacca"	"Carex pilosa"
## [63]	"Carex remota"	"Carex sylvatica"
## [65]	"Centaurium erythraea"	"Cephalanthera damasonium"
## [67]	"Cephalanthera longifolia"	"Cephalanthera rubra"
## [69]	"Cerioporus varius"	"Chlorociboria aeruginascens"
## [71]	"Chondrostereum purpureum"	"Circaea lutetiana"
## [73]	"Cirsium arvense"	"Clematis vitalba"
## [75]	"Clinopodium menthifolium"	"Clinopodium nepeta"
## [77]	"Clinopodium vulgare"	"Coniophora sp."
## [79]	"Crataegus monogyna"	"Cruciata glabra"
## [81]	"Cruciata laevipes"	"Crustoderma sp."
## [83]	"Cyclamen repandum"	"Cymbalaria muralis"
## [85]	"Cynosurus cristatus"	"Cystopteris fragilis"
## [87]	"Cytisophyllum sessilifolium"	"Cytisus scoparius"
## [89]	"Cytisus villosus"	"Dactylis glomerata"
## [91]	"Dactylorhiza maculata"	"Daphne laureola"
## [93]	"Daphne mezereum"	"Dianthus sylvestris"
## [95]	"Diatrype disciformis"	"Diatrype stigma"
## [97]	"Digitalis micrantha"	"Doronicum columnae"
## [99]	"Drymochloa sylvatica"	"Dryopteris filix-mas"
## [101]	"Epilobium montanum"	"Epipactis helleborine"
## [103]	"Epipactis microphylla"	"Epitrichys italica"
## [105]	"Euonymus europaeus"	"Euphorbia amygdaloides"
## [107]	"Euphorbia cyparissias"	"Euphrasia stricta"
## [109]	"Fagus sylvatica"	"Festuca heterophylla"
## [111]	"Fomes fomentarius"	"Fomitopsis pinicola"
## [113]	"Fragaria vesca"	"Fraxinus excelsior"
## [115]	"Galerina marginata"	"Galium aparine"
## [117]	"Galium lucidum"	"Galium odoratum"
## [119]	"Galium rotundifolium"	"Ganoderma applanatum"

## [121]	"Geranium nodosum"	"Geranium purpureum"
## [123]	"Geranium robertianum"	"Glaucumaria carpineae"
## [125]	"Grafia golaka"	"Gyalecta sp."
## [127]	"Gyalolechia flavorubescens"	"Hedera helix"
## [129]	"Helianthemum nummularium"	"Helleborus viridis"
## [131]	"Helminthotheca echioides"	"Hepatica nobilis"
## [133]	"Heterodermia speciosa"	"Hieracium murorum"
## [135]	"Hordelymus europaeus"	"Hypericum montanum"
## [137]	"Hypericum perforatum"	"Hypoxylon fragiforme"
## [139]	"Hypoxylon sp."	"Ilex aquifolium"
## [141]	"Juniperus communis"	"Laburnum anagyroides"
## [143]	"Lachnum relacinum"	"Lapsana communis"
## [145]	"Laserpitium latifolium"	"Lathyrus pratensis"
## [147]	"Lathyrus vernus"	"Lecania naegelii"
## [149]	"Lecanora albella"	"Lecanora argentata"
## [151]	"Lecanora chlorotera"	"Lecanora intumescens"
## [153]	"Lecidella elaeochroma"	"Lenzites betulinus"
## [155]	"Lepra amara"	"Lepraria rigidula"
## [157]	"Lepraria sp."	"Lilium bulbiferum"
## [159]	"Lilium martagon"	"Limodorum abortivum"
## [161]	"Linum viscosum"	"Lonicera alpigena"
## [163]	"Lonicera etrusca"	"Lotus corniculatus"
## [165]	"Luzula forsteri"	"Luzula multiflora"
## [167]	"Luzula sylvatica"	"Medicago lupulina"
## [169]	"Megacollybia platyphylla"	"Melanelixia glabrata"
## [171]	"Melica uniflora"	"Milium effusum"
## [173]	"Moehringia trinervia"	"Monotropa hypophaea"
## [175]	"Mycelis muralis"	"Mycena renati"
## [177]	"Nemania serpens"	"Neottia nidus-avis"
## [179]	"Ochrolechia pallescens"	"Opegrapha sp."
## [181]	"Orthilia secunda"	"Ostrya carpinifolia"
## [183]	"Oxalis acetosella"	"Parmelia sulcata"
## [185]	"Parmelina spp."	"Pertusaria coccodes"
## [187]	"Pertusaria flavida"	"Pertusaria leioplaca"
## [189]	"Pertusaria pertusa"	"Petrosedum rupestre"
## [191]	"Phaeophyscia insignis"	"Phlyctis agelaea"
## [193]	"Phlyctis argena"	"Physcia adscendens"
## [195]	"Physcia airolia"	"Physcia biziana"
## [197]	"Physcia leptalea"	"Physcia sp."
## [199]	"Physcia stellaris"	"Physconia distorta"
## [201]	"Picris hieracioides"	"Platanthera bifolia"
## [203]	"Pleurosticta acetabulum"	"Pleurotus cornucopiae"
## [205]	"Pluteus cervinus"	"Poa compressa"
## [207]	"Poa nemoralis"	"Poa trivialis"
## [209]	"Polygonatum odoratum"	"Polyozosia hagenii"
## [211]	"Polyporus tuberaster"	"Polystichum aculeatum"
## [213]	"Polystichum setiferum"	"Potentilla micrantha"
## [215]	"Prenanthes purpurea"	"Primula vulgaris"
## [217]	"Prunella vulgaris"	"Prunus avium"
## [219]	"Pteridium aquilinum"	"Ptilostemon strictus"
## [221]	"Pulmonaria hirta"	"Ramalina farinacea"
## [223]	"Ramalina fraxinea"	"Ranunculus lanuginosus"
## [225]	"Rosa arvensis"	"Rosa pendulina"
## [227]	"Rubus hirtus group"	"Rubus idaeus"

## [229] "Rumex acetosa"	"Ruscus hypoglossum"
## [231] "Salix caprea"	"Sambucus ebulus"
## [233] "Sanicula europaea"	"Saxifraga paniculata"
## [235] "Saxifraga rotundifolia"	"Schizopora paradoxa"
## [237] "Scleroderma verrucosum"	"Scoliciosporum chlorococcum"
## [239] "Scutellinia sp."	"Sedum cepaea"
## [241] "Senecio nemorensis"	"Sesleria autumnalis"
## [243] "Sesleria nitida"	"Silene italica"
## [245] "Solidago virgaurea"	"Sorbus aria"
## [247] "Sorbus aucuparia"	"Stachys sylvatica"
## [249] "Steccherinum robustius"	"Stereum hirsutum"
## [251] "Stereum sp."	"Stereum subtomentosum"
## [253] "Tanacetum corymbosum"	"Taxus baccata"
## [255] "Tephromela atra"	"Thesium sp"
## [257] "Trametes gibbosa"	"Trametes hirsuta"
## [259] "Trametes ochracea"	"Trametes sp."
## [261] "Trametes versicolor"	"Trifolium campestre"
## [263] "Trifolium ochroleucon"	"Trifolium pratense"
## [265] "Trifolium repens"	"Urtica dioica"
## [267] "Ustulina deusta"	"Verbascum sp"
## [269] "Veronica chamaedrys"	"Veronica cymbalaria"
## [271] "Veronica hederifolia"	"Veronica montana"
## [273] "Veronica officinalis"	"Vicia cracca"
## [275] "Vicia grandiflora"	"Vicia lathyroides"
## [277] "Viola alba"	"Viola reichenbachiana"
## [279] "Xanthoria parietina"	"Xerocomus subtomentosus"
## [281] "Xylaria carpophila"	"Xylaria hypoxylon"
## [283] "Xylaria longipes"	"Xylaria polymorpha"
## [285] "Xylaria sp."	"Xylariaceae sp."

```
#entity richness
```

```
data |>
  group_by(taxon) |>
  summarise(n_distinct(species), .groups = "drop")
```

```
## # A tibble: 3 x 2
##   taxon          'n_distinct(species)'  
##   <fct>                <int>  
## 1 Vascular plants      191  
## 2 Epiphytic lichens    49  
## 3 Saproxylic fungi     46
```

```
#distinct genera
```

```
distinct(data, genus) |> pull() |> sort()
```

## [1] "Abies"	"Acer"	"Actaea"	"Adenostyles"
## [5] "Adoxa"	"Aegopodium"	"Agrimonia"	"Agrostis"
## [9] "Ajuga"	"Allium"	"Anaptychia"	"Aquilegia"
## [13] "Arabis"	"Arctium"	"Aremonia"	"Armillaria"
## [17] "Arthonia"	"Arum"	"Asperula"	"Asplenium"
## [21] "Astragalus"	"Atadinus"	"Athallia"	"Atropa"
## [25] "Avenella"	"Bertia"	"Biscogniauxia"	"Bjerkandera"
## [29] "Brachypodium"	"Bromopsis"	"Buellia"	"Bupleurum"

## [33]	"Caloplaca"	"Calycina"	"Campanula"	"Candelariella"
## [37]	"Cardamine"	"Carduus"	"Carex"	"Centaurium"
## [41]	"Cephalanthera"	"Cerioporus"	"Chlorociboria"	"Chondrostereum"
## [45]	"Circaea"	"Cirsium"	"Clematis"	"Clinopodium"
## [49]	"Coniophora"	"Crataegus"	"Cruciata"	"Crustoderma"
## [53]	"Cyclamen"	"Cymbalaria"	"Cynosurus"	"Cystopteris"
## [57]	"Cytisophyllum"	"Cytisus"	"Dactylis"	"Dactylorhiza"
## [61]	"Daphne"	"Dianthus"	"Diatrype"	"Digitalis"
## [65]	"Doronicum"	"Drymochloa"	"Dryopteris"	"Epilobium"
## [69]	"Epipactis"	"Epitrachys"	"Euonymus"	"Euphorbia"
## [73]	"Euphrasia"	"Fagus"	"Festuca"	"Fomes"
## [77]	"Fomitopsis"	"Fragaria"	"Fraxinus"	"Galerina"
## [81]	"Galium"	"Ganoderma"	"Geranium"	"Glaucumaria"
## [85]	"Grafia"	"Gyalecta"	"Gyalolechia"	"Hedera"
## [89]	"Helianthemum"	"Helleborus"	"Helminthotheca"	"Hepatica"
## [93]	"Heterodermia"	"Hieracium"	"Hordelymus"	"Hypericum"
## [97]	"Hypoxylon"	"Ilex"	"Juniperus"	"Laburnum"
## [101]	"Lachnum"	"Lapsana"	"Laserpitium"	"Lathyrus"
## [105]	"Lecania"	"Lecanora"	"Lecidella"	"Lenzites"
## [109]	"Lepra"	"Lepraria"	"Lilium"	"Limodorum"
## [113]	"Linum"	"Lonicera"	"Lotus"	"Luzula"
## [117]	"Medicago"	"Megacollybia"	"Melanelixia"	"Melica"
## [121]	"Miliun"	"Moehringia"	"Monotropa"	"Mycelis"
## [125]	"Mycena"	"Nemania"	"Neottia"	"Ochrolechia"
## [129]	"Opegrapha"	"Orthilia"	"Ostrya"	"Oxalis"
## [133]	"Parmelia"	"Parmelina"	"Pertusaria"	"Petrosedum"
## [137]	"Phaeophyscia"	"Phlyctis"	"Physcia"	"Physconia"
## [141]	"Picris"	"Platanthera"	"Pleurosticta"	"Pleurotus"
## [145]	"Pluteus"	"Poa"	"Polygonatum"	"Polyozosia"
## [149]	"Polyporus"	"Polystichum"	"Potentilla"	"Prenanthes"
## [153]	"Primula"	"Prunella"	"Prunus"	"Pteridium"
## [157]	"Ptilostemon"	"Pulmonaria"	"Ramalina"	"Ranunculus"
## [161]	"Rosa"	"Rubus"	"Rumex"	"Ruscus"
## [165]	"Salix"	"Sambucus"	"Sanicula"	"Saxifraga"
## [169]	"Schizopora"	"Scleroderma"	"Scoliciosporum"	"Scutellinia"
## [173]	"Sedum"	"Senecio"	"Sesleria"	"Silene"
## [177]	"Solidago"	"Sorbus"	"Stachys"	"Steccherinum"
## [181]	"Stereum"	"Tanacetum"	"Taxus"	"Tephromela"
## [185]	"Thesium"	"Trametes"	"Trifolium"	"Urtica"
## [189]	"Ustulina"	"Verbascum"	"Veronica"	"Vicia"
## [193]	"Viola"	"Xanthoria"	"Xerocomus"	"Xylaria"
## [197]	"Xylariaceae"			

```
#genus richness
data |>
  #Xylariaceae is not a genus, don't count
  filter_out(genus == "Xylariaceae") |>
  group_by(taxon) |>
  summarise(n_distinct(genus), .groups = "drop")
```

```
## # A tibble: 3 x 2
##   taxon          'n_distinct(genus)'
##   <fct>          <int>
## 1 Vascular plants      132
```

```
## 2 Epiphytic lichens      31
## 3 Saproxylic fungi      33
```

```
#distinct epithets
```

```
distinct(data, epit) |> pull() |> sort()
```

```
## [1] "abortivum"      "acetabulum"    "acetosa"       "acetosella"
## [5] "aculeatum"      "adscendens"    "adusta"        "aeruginascens"
## [9] "agelaea"        "agrimonoides"  "aipolia"       "alba"
## [13] "albella"        "alpigena"      "alpina"        "alpinus"
## [17] "amara"          "amygdaloides"  "anagyroides"   "aparine"
## [21] "applanatum"     "aquifolium"    "aquilinum"     "argena"
## [25] "argentata"      "aria"          "arvense"       "arvensis"
## [29] "atra"          "aucuparia"     "australis"     "autumnalis"
## [33] "avium"         "baccata"       "bella-donna"   "betulinus"
## [37] "bifolia"        "biziana"       "bulbifera"     "bulbiferum"
## [41] "campestre"      "capillaris"    "caprea"        "carpineae"
## [45] "carpinifolia"   "carpophila"    "cepaea"        "cerina"
## [49] "cerinelloides" "cervinus"      "chamaedrys"    "chlarotera"
## [53] "chlorococcum"   "ciliaris"      "citrina"       "coccodes"
## [57] "collina"        "columnae"      "communis"      "compressa"
## [61] "corniculatus"   "cornucopiae"   "corymbosum"    "cracca"
## [65] "crinalis"       "cristatus"     "cymbalaria"    "cyparissias"
## [69] "damasonium"     "deusta"        "didyma"        "digitata"
## [73] "dioica"         "disciformis"   "distorta"      "divulsa"
## [77] "dumeticola"     "ebulus"        "echioides"     "effusum"
## [81] "elaeochroma"    "enneaphyllos"  "erythraea"     "etrusca"
## [85] "eupatoria"      "europaea"      "europaeus"     "excelsior"
## [89] "faginea"        "falcatum"      "farinacea"     "filix-mas"
## [93] "flacca"         "flavida"       "flavorubescens" "flexuosa"
## [97] "fomentarius"    "forsteri"      "fragiforme"    "fragilis"
## [101] "fraxinea"       "gibbosa"       "glabra"        "glabratula"
## [105] "glomerata"      "glycyphyllos"  "golaka"        "graeca"
## [109] "grandiflora"    "griseovirens"  "hagenii"       "hederifolia"
## [113] "helix"          "helleborine"   "heptaphylla"   "heterophylla"
## [117] "hieracioides"   "hirsuta"       "hirsutum"      "hirta"
## [121] "hirtus"         "hypoglossum"   "hypophegea"    "hypoxylon"
## [125] "idaeus"         "insignis"      "intumescens"   "italica"
## [129] "kitaibelii"     "laevigata"     "laevipes"      "lanuginosus"
## [133] "lappa"          "lathyroides"   "latifolium"    "laureola"
## [137] "leioplaca"      "leptalea"      "longifolia"    "longipes"
## [141] "lucidum"        "lupulina"      "lutetiana"     "maculata"
## [145] "maculatum"      "marginata"     "martagon"      "menthifolium"
## [149] "mezereum"       "micrantha"     "microphylla"   "minus"
## [153] "monogyna"       "montana"       "montanum"      "moriformis"
## [157] "moschatellina"  "multiflora"    "muralis"       "murorum"
## [161] "naegelii"       "nemoralis"     "nemorensis"    "nepeta"
## [165] "nidus-avis"     "nitida"        "nobilis"       "nodosum"
## [169] "nummularia"     "nummularium"   "ochracea"      "ochroleucon"
## [173] "odoratum"       "officinalis"   "opalus"        "pallescens"
## [177] "paniculata"     "paradoxa"      "parietina"     "pendulina"
## [181] "perforatum"     "persicifolia"  "pertusa"       "pilosa"
## [185] "pinicola"       "platanoides"   "platyphylla"   "podagraria"
## [189] "polymorpha"     "pratense"      "pratensis"     "pseudoplatanus"
```

```
## [193] "purpurea"      "purpureum"      "pyracea"        "radiata"
## [197] "ramosa"        "rapunculus"     "reichenbachiana" "relicinum"
## [201] "remota"        "renati"         "repandum"       "repens"
## [205] "reptans"       "rigidula"       "robertianum"    "robustus"
## [209] "rotundifolia"  "rotundifolium"  "rubra"          "rupestre"
## [213] "ruta-muraria"  "scoparius"      "secunda"        "serpens"
## [217] "sessilifolium" "setiferum"      "sp"             "sp."
## [221] "speciosa"      "spicata"         "spp."           "stellaris"
## [225] "stigma"        "stricta"        "strictus"       "subtomentosum"
## [229] "subtomentosus" "sulcata"        "sylvatica"      "sylvaticum"
## [233] "sylvestris"    "tanfanii"       "trachelium"     "trichomanes"
## [237] "trinervia"     "trivialis"      "tuberaster"     "uniflora"
## [241] "ursinum"       "varius"         "vernus"         "verrucosum"
## [245] "versicolor"    "vesca"          "villosus"       "virgaurea"
## [249] "viridis"       "viscosum"       "vitalba"        "vulgare"
## [253] "vulgaris"
```

```
#check incerta
incerta_index <- c("", "sp.", "spp.")

incerta <- data |>
  filter(epit %in% incerta_index) |>
  distinct(taxon, species, genus)

incerta
```

```
## # A tibble: 15 x 3
##   taxon          species      genus
##   <fct>         <chr>      <chr>
## 1 Epiphytic lichens Caloplaca sp. Caloplaca
## 2 Epiphytic lichens Gyalecta sp. Gyalecta
## 3 Epiphytic lichens Lepraria sp. Lepraria
## 4 Epiphytic lichens Opegrapha sp. Opegrapha
## 5 Epiphytic lichens Parmelina spp. Parmelina
## 6 Epiphytic lichens Physcia sp. Physcia
## 7 Saproxylic fungi Armillaria sp. Armillaria
## 8 Saproxylic fungi Coniophora sp. Coniophora
## 9 Saproxylic fungi Crustoderma sp. Crustoderma
## 10 Saproxylic fungi Hypoxylon sp. Hypoxylon
## 11 Saproxylic fungi Scutellinia sp. Scutellinia
## 12 Saproxylic fungi Stereum sp. Stereum
## 13 Saproxylic fungi Trametes sp. Trametes
## 14 Saproxylic fungi Xylaria sp. Xylaria
## 15 Saproxylic fungi Xylariaceae sp. Xylariaceae
```

```
#check congenetics of incerta

data |>
  filter(genus %in% incerta$genus) |>
  distinct(species, taxon) |>
  print(n = 50)
```

```
## # A tibble: 33 x 2
```



```
## species taxon
## <chr> <fct>
## 1 Caloplaca cerina Epiphytic lichens
## 2 Caloplaca sp. Epiphytic lichens
## 3 Gyalecta sp. Epiphytic lichens
## 4 Lepraria rigidula Epiphytic lichens
## 5 Lepraria sp. Epiphytic lichens
## 6 Opeggrapha sp. Epiphytic lichens
## 7 Parmelina spp. Epiphytic lichens
## 8 Physcia adscendens Epiphytic lichens
## 9 Physcia aipolia Epiphytic lichens
## 10 Physcia biziana Epiphytic lichens
## 11 Physcia leptalea Epiphytic lichens
## 12 Physcia sp. Epiphytic lichens
## 13 Physcia stellaris Epiphytic lichens
## 14 Armillaria sp. Saproxylic fungi
## 15 Coniophora sp. Saproxylic fungi
## 16 Crustoderma sp. Saproxylic fungi
## 17 Hypoxylon fragiforme Saproxylic fungi
## 18 Hypoxylon sp. Saproxylic fungi
## 19 Scutellinia sp. Saproxylic fungi
## 20 Stereum hirsutum Saproxylic fungi
## 21 Stereum sp. Saproxylic fungi
## 22 Stereum subtomentosum Saproxylic fungi
## 23 Trametes gibbosa Saproxylic fungi
## 24 Trametes hirsuta Saproxylic fungi
## 25 Trametes ochracea Saproxylic fungi
## 26 Trametes sp. Saproxylic fungi
## 27 Trametes versicolor Saproxylic fungi
## 28 Xylaria carpophila Saproxylic fungi
## 29 Xylaria hypoxylon Saproxylic fungi
## 30 Xylaria longipes Saproxylic fungi
## 31 Xylaria polymorpha Saproxylic fungi
## 32 Xylaria sp. Saproxylic fungi
## 33 Xylariaceae sp. Saproxylic fungi
```

#those that have no congenetics

```
keep_genus <- data |>
  filter(genus %in% incerta$genus) |>
  distinct(genus, species, taxon) |>
  add_count(genus) |>
  filter(n == 1)
```

```
keep_genus
```

```
## # A tibble: 8 x 4
## genus species taxon n
## <chr> <chr> <fct> <int>
## 1 Gyalecta Gyalecta sp. Epiphytic lichens 1
## 2 Opeggrapha Opeggrapha sp. Epiphytic lichens 1
## 3 Parmelina Parmelina spp. Epiphytic lichens 1
## 4 Armillaria Armillaria sp. Saproxylic fungi 1
## 5 Coniophora Coniophora sp. Saproxylic fungi 1
```

```
## 6 Crustoderma Crustoderma sp. Saproxylic fungi      1
## 7 Scutellinia Scutellinia sp. Saproxylic fungi      1
## 8 Xylariaceae Xylariaceae sp. Saproxylic fungi      1
```

These should be kept because they are the only ones in their genus. Now, let's check the ones with more entries:

```
data |>
  filter(genus %in% incerta$genus) |>
  distinct(genus, species, taxon) |>
  add_count(genus) |>
  filter(n > 1) |>
  print(n = 50)
```

```
## # A tibble: 25 x 4
##   genus      species      taxon      n
##   <chr>    <chr>      <fct>    <int>
## 1 Caloplaca Caloplaca cerina Epiphytic lichens 2
## 2 Caloplaca Caloplaca sp.   Epiphytic lichens 2
## 3 Lepraria  Lepraria rigidula Epiphytic lichens 2
## 4 Lepraria  Lepraria sp.      Epiphytic lichens 2
## 5 Physcia   Physcia adscendens Epiphytic lichens 6
## 6 Physcia   Physcia aipolia   Epiphytic lichens 6
## 7 Physcia   Physcia biziana   Epiphytic lichens 6
## 8 Physcia   Physcia leptalea  Epiphytic lichens 6
## 9 Physcia   Physcia sp.       Epiphytic lichens 6
## 10 Physcia   Physcia stellaris Epiphytic lichens 6
## 11 Hypoxylon Hypoxylon fragiforme Saproxylic fungi 2
## 12 Hypoxylon Hypoxylon sp.      Saproxylic fungi 2
## 13 Stereum   Stereum hirsutum   Saproxylic fungi 3
## 14 Stereum   Stereum sp.        Saproxylic fungi 3
## 15 Stereum   Stereum subtomentosum Saproxylic fungi 3
## 16 Trametes  Trametes gibbosa   Saproxylic fungi 5
## 17 Trametes  Trametes hirsuta   Saproxylic fungi 5
## 18 Trametes  Trametes ochracea  Saproxylic fungi 5
## 19 Trametes  Trametes sp.       Saproxylic fungi 5
## 20 Trametes  Trametes versicolor Saproxylic fungi 5
## 21 Xylaria   Xylaria carpophila Saproxylic fungi 5
## 22 Xylaria   Xylaria hypoxylon  Saproxylic fungi 5
## 23 Xylaria   Xylaria longipes   Saproxylic fungi 5
## 24 Xylaria   Xylaria polymorpha Saproxylic fungi 5
## 25 Xylaria   Xylaria sp.        Saproxylic fungi 5
```

Of those, *Lepraria* is kept because it was different from *Lepraria rigidula*, while *Physcia* and *Caloplaca* should be removed.

```
keep_genus <- c(keep_genus$genus, "Lepraria")

#keep only those incerta
incerta <- incerta |>
  filter(!(genus %in% keep_genus))

data <- data |>
  anti_join(incerta)
```

```
## Joining with 'by = join_by(taxon, species, genus)'
```

```
datal <- datal |>
  anti_join(incerta)
```

```
## Joining with 'by = join_by(taxon, species, genus)'
```

```
# mean richness per area -----
```

```
data |>
  summarise(richness = n_distinct(species), .by = c(taxon, site, plot)) |>
  summarise(avg = mean(richness), .by = c(taxon, site)) |>
  mutate(avg = round(avg))
```

```
## # A tibble: 9 x 3
##   taxon          site      avg
##   <fct>         <fct>    <dbl>
## 1 Vascular plants Incodaro      31
## 2 Vascular plants Venaquaro     56
## 3 Vascular plants Prati di Tivo  66
## 4 Epiphytic lichens Prati di Tivo  16
## 5 Epiphytic lichens Incodaro     14
## 6 Epiphytic lichens Venaquaro     12
## 7 Saproxylic fungi Venaquaro       4
## 8 Saproxylic fungi Incodaro       5
## 9 Saproxylic fungi Prati di Tivo    7
```

```
# mean, min, max per layer and area -----
```

```
datal |>
  summarise(richness = n_distinct(species),
    .by = c(taxon, site, plot)) |>
  summarise(avg = mean(richness),
    min = min(richness),
    max = max(richness),
    .by = c(taxon, site)) |>
  mutate(avg = round(avg)) |>
  arrange(taxon, site)
```

```
## # A tibble: 15 x 5
##   taxon          site      avg  min  max
##   <fct>         <fct>    <dbl> <int> <int>
## 1 Tree          Incodaro      2     2    3
## 2 Tree          Prati di Tivo  2     1    2
## 3 Tree          Venaquaro      2     1    3
## 4 Shrub         Incodaro      2     1    2
## 5 Shrub         Prati di Tivo  2     1    3
## 6 Shrub         Venaquaro      2     1    3
## 7 Herb          Incodaro     31    18   45
## 8 Herb          Prati di Tivo  66    57   79
## 9 Herb          Venaquaro     56    47   75
## 10 Epiphytic lichens Incodaro     14     9   20
```

```
## 11 Epiphytic lichens Prati di Tivo    16     9    21
## 12 Epiphytic lichens Venaquaro       12     8    19
## 13 Saproxylic fungi  Incodaro         5     2    11
## 14 Saproxylic fungi  Prati di Tivo    7     2    16
## 15 Saproxylic fungi  Venaquaro        4     1     8
```

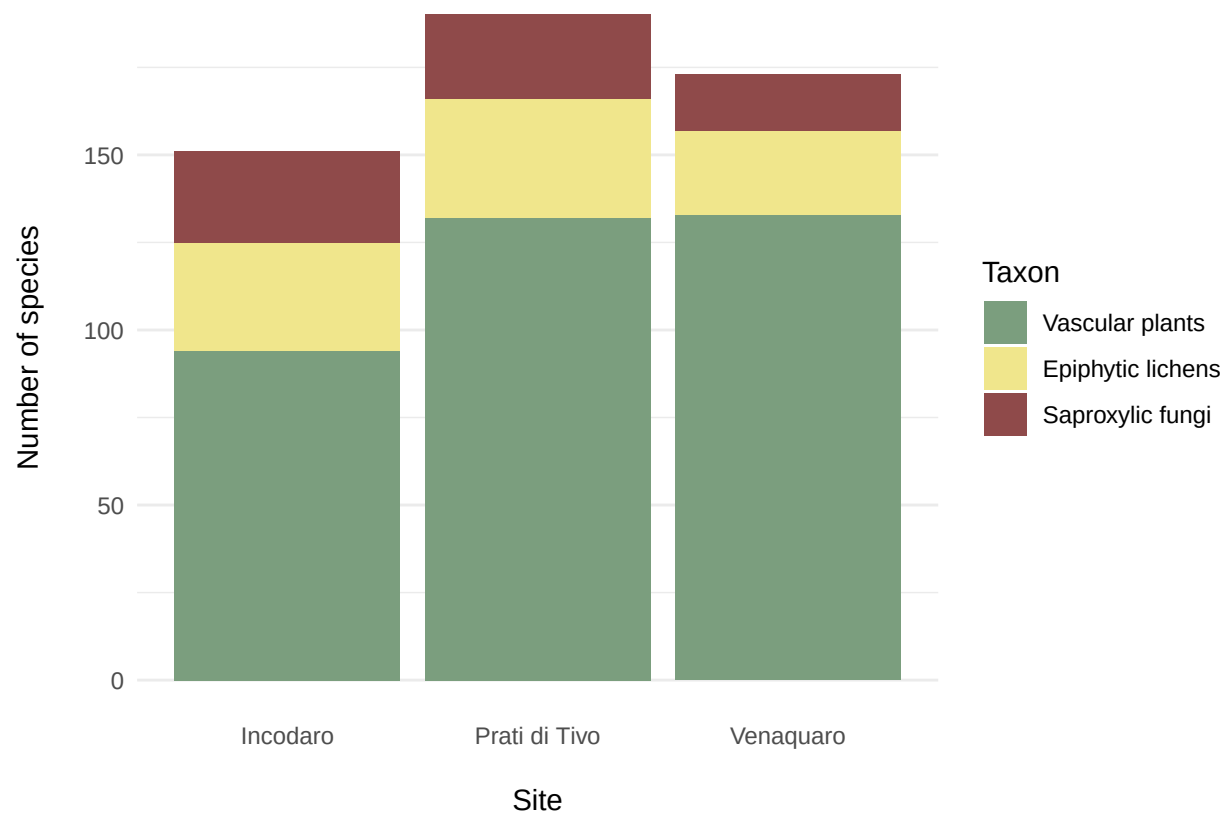
```
# plot! -----
```

```
richness <- data |>
  group_by(site, taxon) |>
  summarise(rich = n_distinct(species), .groups = "drop")

richness |>
  arrange(taxon, desc(rich))
```

```
## # A tibble: 9 x 3
##   site      taxon      rich
##   <fct>    <fct>    <int>
## 1 Venaquaro Vascular plants  133
## 2 Prati di Tivo Vascular plants  132
## 3 Incodaro   Vascular plants   94
## 4 Prati di Tivo Epiphytic lichens  34
## 5 Incodaro   Epiphytic lichens  31
## 6 Venaquaro   Epiphytic lichens  24
## 7 Incodaro   Saproxylic fungi   26
## 8 Prati di Tivo Saproxylic fungi   24
## 9 Venaquaro   Saproxylic fungi   16
```

```
richness |>
  ggplot(aes(x = site, y = rich, fill = taxon)) +
  geom_col(position = position_stack(reverse = TRUE)) +
  scale_fill_discrete(palette = c("#7B9E7E", "#F0E68C", "#8F4A4A")) +
  theme_minimal() +
  labs(
    x = "Site",
    y = "Number of species",
    fill = "Taxon"
  ) +
  theme(
    axis.title.x = element_text(margin = margin(t = 15)),
    axis.title.y = element_text(margin = margin(r = 15)),
    plot.title = element_text(hjust = 0.5, size = 16, face = "bold", margin = margin(b = 15)),
    panel.grid.major.x = element_blank()
  )
```



```
ggsave("richness_stack.png", width = 119, height = 73, scale = 1.2, units = "mm", bg = "white")  
ggsave("Fig2.eps", width = 119, height = 73, scale = 1.2, units = "mm", bg = "white")
```